

MUHAMMAD ADNAN ANWAR, PH.D.

Data Engineer | Python • SQL • Data Pipelines • ETL

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SUMMARY

Data Engineer with a Ph.D. in Computational Mathematics and strong experience in building data pipelines, processing large-scale datasets, and developing data-driven systems. Skilled in Python, SQL, and ETL workflows, with hands-on experience in data ingestion, transformation, and pipeline automation. Experienced in working with structured and unstructured data, including scientific and real-world datasets. Currently focused on building scalable data pipelines, workflow orchestration, and cloud-based data systems.

TECHNICAL SKILLS

Data Engineering

- ETL/ELT pipeline development
- Data ingestion, cleaning, and transformation
- Batch data processing workflows
- Data modeling and schema design

Programming & Querying

- Python (Pandas, NumPy, data processing workflows)
- SQL (PostgreSQL, MySQL – joins, CTEs, query optimization)
- Bash, Git

Tools & Platforms

- Workflow concepts (Airflow – learning/implementation level)
- Data storage: PostgreSQL, file-based systems (CSV, JSON, NumPy)
- Vector databases: FAISS
- API development and data interfaces (REST, Streamlit)

Analytics & Visualization

- Power BI, Tableau
- Excel (advanced: Power Query, dashboards, automation)

PROFESSIONAL EXPERIENCE & PROJECTS

Data Pipeline for Retrieval-Augmented Systems

- Developed structured data ingestion and transformation workflows for document-based systems
- Automated preprocessing for unstructured datasets (PDF, JSON)
- Designed scalable retrieval pipeline using embeddings and semantic search
- Improved data quality and retrieval accuracy through pipeline optimization

Customer Analytics Pipeline (Retail Banking – DataThon Project)

- Built ETL pipeline for cleaning, transforming, and preparing customer data
- Processed structured datasets for downstream analytics and modeling
- Designed reproducible workflows for feature engineering and data preparation
- Optimized SQL queries for efficient data extraction and analysis
- Delivered end-to-end analytical solution in a team environment (Runner-Up position)

Scientific Data Processing Pipeline (4D MRI Flow Data)

- Developed data processing workflows for high-dimensional scientific datasets
- Handled large-scale structured data and extracted meaningful patterns
- Integrated machine learning models into data processing pipelines

EDUCATION

Ph.D. in Computational Mathematics

Lahore University of Management Sciences (LUMS), Pakistan — 2022

M.Phil. in Applied Mathematics — 2018

University of Engineering and Technology, Lahore, Pakistan

B.S. in Mathematics (Minor in Computer Science) — 2016

Bahauddin Zakariya University, Multan, Pakistan

TEACHING & LEADERSHIP

Data Science & Python Instructor

- Taught SQL, Python, and data analysis concepts with practical datasets
- Designed hands-on exercises focused on real-world data problems
- Delivered structured learning materials for analytical and programming skills

PROFESSIONAL DEVELOPMENT

- Advanced SQL & Data Analytics (365 Careers)
- Python for Data Science (Udemy)
- Power BI Dashboard Development
- Introduction to Deep Learning (MIT – selected modules)
- Responsible AI & Model Interpretability (SHAP, LIME concepts)

ADDITIONAL INFORMATION

- Experience working with sensitive scientific and structured datasets (awareness of data privacy principles)
- Strong analytical thinking and problem-solving skills
- Fluent in English